

PREFERENCES—WARM-UP PROBLEM SET

ECON101—INTRODUCTION TO MICROECONOMICS.

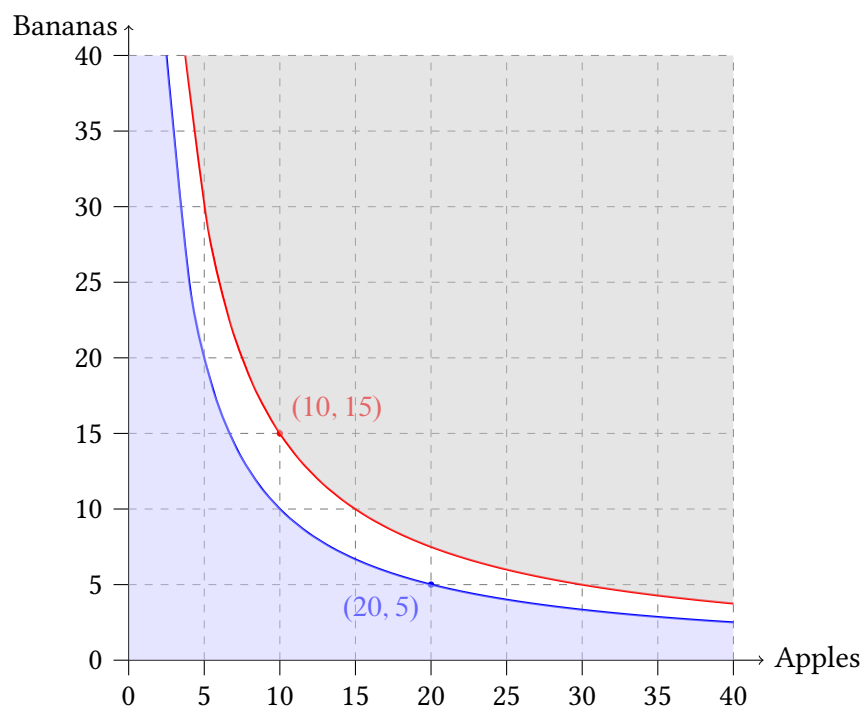
SPRING 2026

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Problem 1

(a) and (b) See the graph below.



(c) $(30, 5) \sim (10, 15)$. True

(d) $(10, 15) \succ (20, 5)$. True

(e) $(20, 5) \succeq (10, 10)$. True

(f) $(24, 4) \succeq (11, 9.1)$. False

(g) $(11, 14) \succ (2, 49)$. True

(h) Yes

(i) No

(j) $MRS = -1$

(k) $MRS = -4$

(l) $MRS = -0.25$

(m) Yes.

Problem 2

(a) and (b) See the graph below

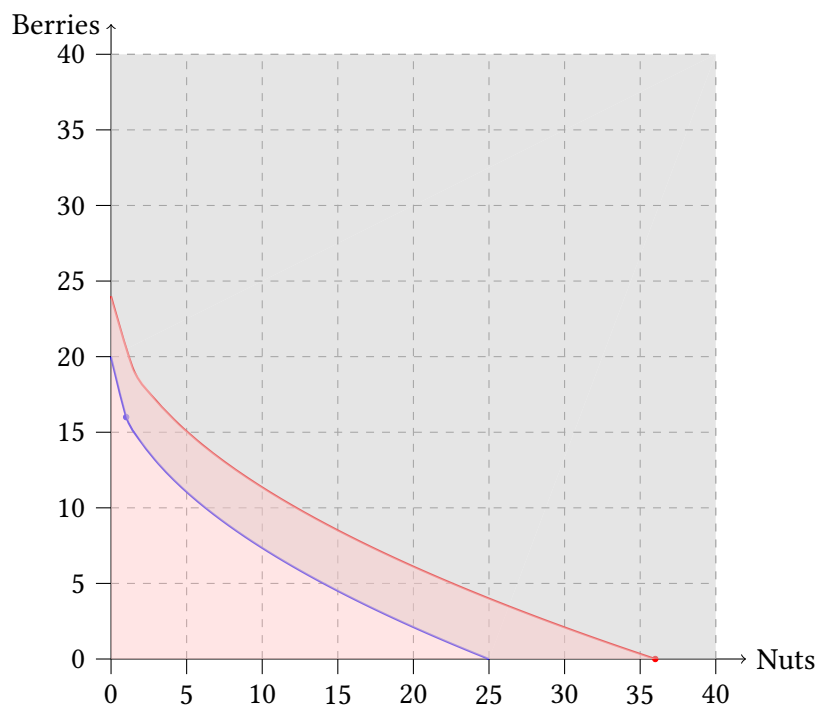
(c) What is the slope of Nazerke's indifference curve at the point (9, 8)? (Hint: Recall from calculus the way to calculate the slope of a curve. If you don't know calculus, you will have to draw your diagram carefully and estimate the slope.) $-2/3$.

(d) What is the slope of her indifference curve at the point (4, 12)? -1 .

(e) What is the slope of her indifference curve at the point (9, 12)? $-2/3$. What about at the point (4, 16)? -1 .

(f) Yes

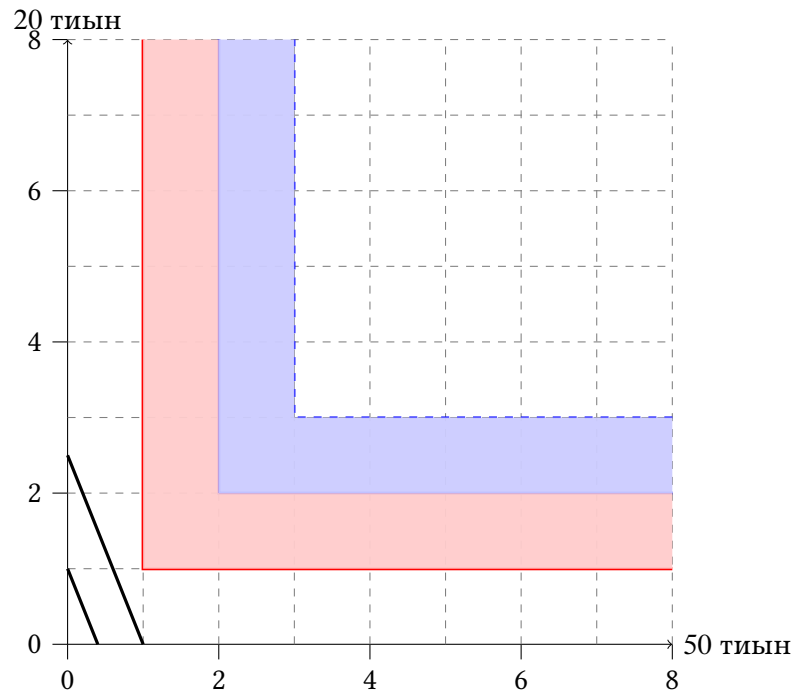
(g) Yes.



Problem 3

(a) 2.

(b) See the graph below.

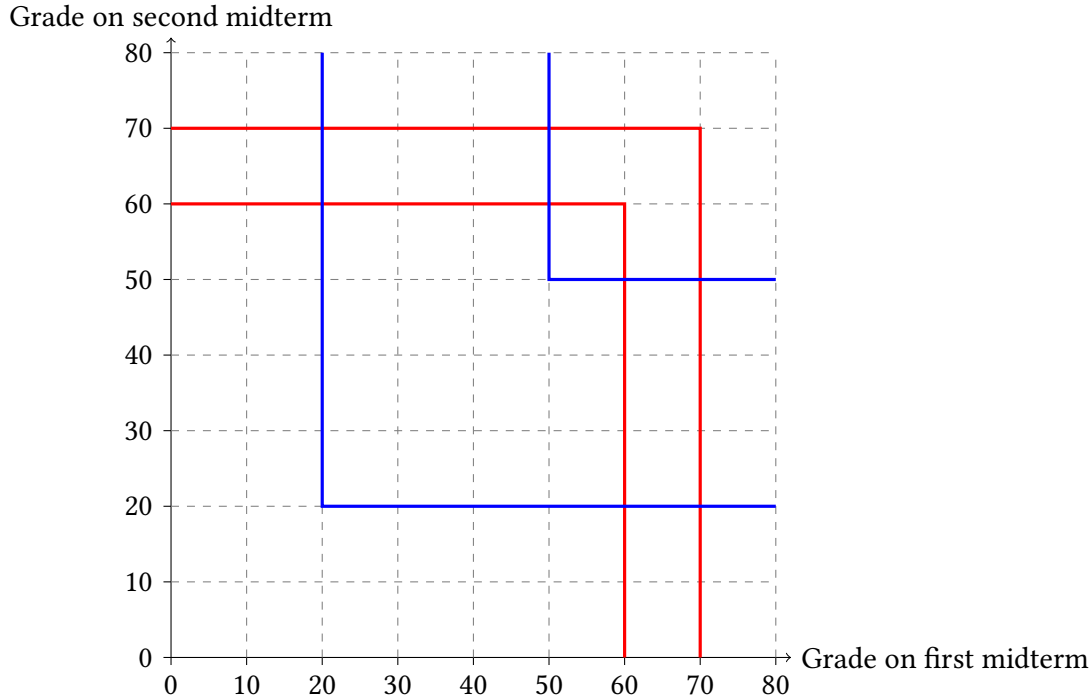


- (c) Yes.
- (d) No.
- (e) No.
- (f) See the graph above. The new indifference curves would be line segments with slope -2.5 , because one 50 ТИЫН coin exchanges for two and a half 20 ТИЫН coins.

Problem 4

Professor Legkovato always gives two midterms in his business class. He only uses the higher of the two scores that a student gets on the midterms when he calculates the course grade.

- (a) Nazima prefers the bundle (20, 70).
- (b) See the graph. below.
- (c) No.



Nazima is also taking a course in economics from Professor AlekHARDov. Professor AlekHARDov gives two midterms. Instead of discarding the lower grade, Professor AlekHARDov discards the higher one.

- (d) Nazima prefers the bundle $(60, 50)$.
- (e) See the graph above.
- (f) Yes.

Problem 5

Coach Anabolinov likes his players to be big, fast, and obedient. If player A is better than player B in two of these three characteristics, then Coach Anabolinov prefers A to B , but if B is better than A in two of these three characteristics, then Anabolinov prefers B to A . Otherwise, Anabolinov is indifferent between them. Wilbur weighs 340 pounds, runs very slowly, and is fairly obedient. Harold weighs 240 pounds, runs very fast, and is very disobedient. Jerry weighs 150 pounds, runs at average speed, and is extremely obedient.

- (a) Anabolinov prefers W to H .
- (b) Anabolinov prefers H to J .
- (c) Anabolinov prefers J to W .
- (d) No: $W \succ H$, and $H \succ J$ but it is not the case that $W \succ J$.
- (e) After several losing seasons, Coach Anabolinov decides to change his way of judging players. According to his new preferences, Anabolinov prefers player A to player B if player A is better

in all three of the characteristics that Anabolinov values, and he prefers B to A if player B is better at all three things. He is indifferent between A and B if they weigh the same, are equally fast, and are equally obedient. In all other cases, Coach Anabolinov simply says “ A and B are not comparable.”

- (f) No, they are not complete.
- (g) Yes, they would be transitive.